



ALGEBRA 2

CLASS 13

# Rational Expressions

Simplifying, Multiplying, and Dividing


$$\frac{P(x)}{Q(x)}$$
$$Q(x) \neq 0$$

*Build confidence. Recognize patterns. Think mathematically.*

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# Understanding Rational Expressions

A **rational expression** is a fraction whose numerator and denominator are polynomials.

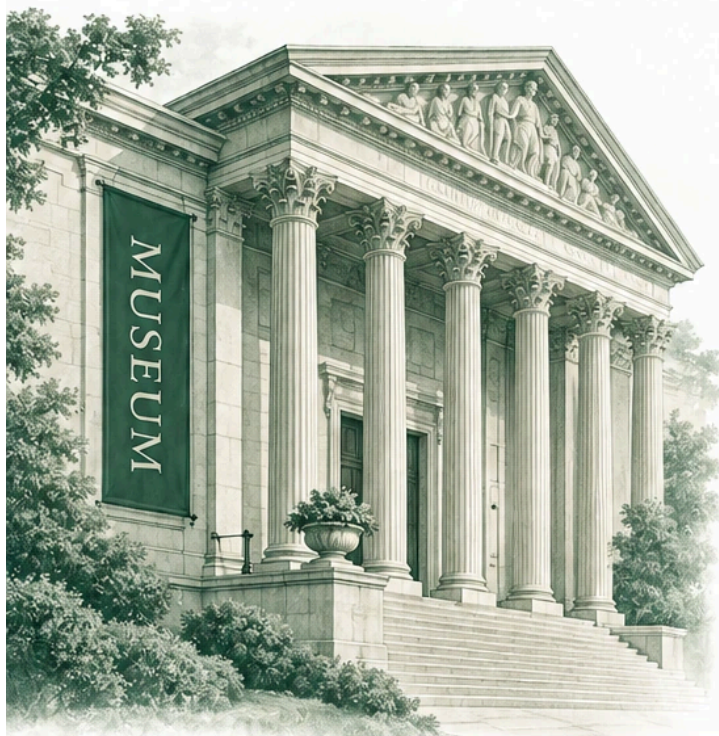
$$\frac{x + 2}{x - 3}$$

**Numerator:**  
polynomial

**Denominator:**  
polynomial

The denominator cannot equal zero.

Here,  $x \neq 3$ .



## Real-world connection: A museum visit

Suppose a group pays a \$120 guide fee plus \$5 per student.

$x$  = number of students.

**Average cost per student =**

$$\frac{120 + 5x}{x}$$

For 20 students:

$$\frac{120 + 5 \cdot 20}{20} = \$11 \text{ per student.}$$

In this context,  $x$  is a positive whole number.



### MATH HINT

The numerator gives the total cost. Dividing by  $x$  gives the cost per student.

## Image 1 practice: Understanding rational expressions

**Q1.** For  $F(x) = \frac{5x - 2}{2x + 6}$ , identify the numerator and denominator, find the excluded value, and evaluate  $F(-2)$ .

**Q2.** Evaluate  $\frac{3x^2 - 5}{2x - 1}$  at  $x = -2$  and  $x = \frac{1}{2}$ . Explain any undefined result.

**Q3.** For  $\frac{x^2 - 9}{x^2 - x - 6}$ , classify the result at  $x = -3, 2, 3$  as zero, nonzero, or undefined. Give each defined value.

**Q4.** Consider  $\frac{3x + 9}{x - 5}$ . (a) Which value of  $x$  makes the expression undefined? (b) Which value makes it equal to zero? (c) Find its value at  $x = 2$ .

**Q5.** A workshop charges \$108 plus \$8 per student. A sponsor pays \$24 of the bill. Write the average amount paid by each of  $n$  students and compare the averages for 7 and 12 students.

# Finding Excluded Values

**Excluded values** make the original denominator equal to zero. They are not allowed.

$$\frac{x + 1}{x^2 - 9}$$

**1** Set the denominator equal to zero.

$$x^2 - 9 = 0$$

**2** Factor.

$$(x - 3)(x + 3) = 0$$

**3** Solve and exclude.

$$x = 3 \text{ or } x = -3$$

**Restrictions:**  $x \neq 3$  and  $x \neq -3$ .

Domain: all real numbers except  $-3$  and  $3$ .



## Real-world connection: Cycling speed

A cyclist travels 12 miles in  $t$  hours.

$$\text{Average speed} = \frac{12}{t} \text{ mph.}$$

Here,  $t \neq 0$ .

Dividing by zero is undefined.

For this trip, elapsed time must be positive:  $t > 0$ .

If  $t = 1.5$ , the average speed is 8 mph.



### MATH HINT

Check the **ORIGINAL** denominator before simplifying.  
A canceled factor does not remove an excluded value.

## Image 2 practice: Finding excluded values

**Q6.** Find all excluded values of  $\frac{x+1}{3x^2-12x}$ . Explain why both factors of the denominator matter.

**Q7.** Find the real domain of  $\frac{x-2}{(2x-3)(x+5)}$ , and decide whether  $x = \frac{3}{2}$  is allowed.

**Q8.** Find every excluded value of  $\frac{x^2-16}{x^2+2x-8}$ . Which exclusion comes from a factor that cancels?

**Q9.** Find the real domain of  $\frac{2x-1}{2x^2-7x+3}$ . Is  $x = \frac{1}{2}$  a zero of the original expression?

**Q10.** Find every excluded value of  $\frac{x^2-9}{2x^3-18x}$ , then simplify. Which restrictions would be missed by looking only at the simplified denominator?

# Factoring Before Simplifying

**Factor** the numerator and denominator completely to reveal common factors.

$$\frac{x^2 + 5x + 6}{x^2 - 4}$$

**1** Factor the numerator.

$$x^2 + 5x + 6 = (x + 2)(x + 3)$$

$2 + 3 = 5$  and  $2 \times 3 = 6$ .

**2** Factor the denominator.

$$x^2 - 4 = (x + 2)(x - 2)$$

Use the difference of squares.

**3** Rewrite using factors.

$$\frac{(x + 2)(x + 3)}{(x + 2)(x - 2)}$$

**Common factor:**  $x + 2$ .

**Original restrictions:**  $x \neq -2$  and  $x \neq 2$ .



## Real-world connection: Garden design

A rectangular garden has area  $x^2 + 5x + 6$  square meters and width  $x + 2$  meters, where  $x > 0$ .

$$\text{Length} = \frac{x^2 + 5x + 6}{x + 2} = x + 3 \text{ meters.}$$

Factoring reveals the missing dimension.



### MATH HINT

Look for a greatest common factor first, then check for trinomials and special patterns.

**Image 3 practice: Factoring before simplifying**

**Q11.** Simplify and state every excluded value:  $\frac{8x^2 + 24x}{4x}$ .

**Q12.** Simplify and state every excluded value:  $\frac{2x^2 - 50}{4x + 20}$ .

**Q13.** Simplify and state every excluded value:  $\frac{2x^2 + 11x + 12}{2x + 3}$ .

**Q14.** Simplify and state every excluded value:  $\frac{x^2 + 2x - 15}{x^2 - 6x + 9}$ .

**Q15.** Simplify and state every excluded value:  $\frac{6x^2 + 3x - 9}{4x^2 + 8x - 12}$ .

# Simplifying Rational Expressions

Cancel common factors, then keep every restriction from the original denominator.

$$\frac{(x+2)(x+3)}{(x+2)(x-2)}$$

**1** Record the restrictions.

$$(x+2)(x-2) \neq 0$$

$$x \neq -2 \text{ and } x \neq 2.$$

**2** Cancel the common factor.

$$\frac{\cancel{(x+2)}(x+3)}{\cancel{(x+2)}(x-2)}$$

$$\frac{x+3}{x-2} = 1 \text{ because } x \neq -2.$$

**3** Write the simplified expression.

$$\frac{x+3}{x-2} \quad x \neq -2, 2$$

The expression changes form; the original restrictions remain.



## Real-world connection: Printing posters

Suppose the total cost of printing  $n$  posters is  $4n$  dollars, plus a setup fee of 20 dollars, where  $n$  is a positive whole number.

$$\text{Average cost} = \frac{4n + 20}{n} = 4 + \frac{20}{n}$$

Only the common factor in  $4n/n$  cancels.  
The setup fee is shared across the order.



### MATH HINT

Cancel factors, not terms. Never cancel across addition or subtraction.

**Image 4 practice: Simplifying and keeping restrictions**

**Q16.** Simplify and state every excluded value:  $\frac{18x^3}{24x^2}$ .

**Q17.** Simplify and state every excluded value:  $\frac{(x-4)^2(x+6)}{(x-4)(x+6)^2}$ .

**Q18.** Simplify and state every excluded value:  $\frac{4x^2 - 49}{2x^2 + 9x + 7}$ .

**Q19.** Simplify and state every excluded value:  $\frac{2x^2 - 12x + 18}{3x^2 - 27}$ .

**Q20.** Simplify and state every excluded value:  $\frac{6x^3 - 24x^2}{3x^3 - 24x^2 + 48x}$ .

# Avoiding Common Cancellation Mistakes

Cancel common factors, never individual terms joined by addition or subtraction.

**✗ INCORRECT**

$$\frac{x + 3}{x} \neq 3$$

The  $x$  in  $x + 3$  is a term, not a factor of the whole numerator.

**✓ CORRECT**

$$\frac{3x}{x} = 3, x \neq 0$$

The numerator is a product:  $3 \times x$ . The common factor  $x$  cancels.

**1 Factor first when possible.**

$$\frac{3x + 6}{3} = \frac{3(x + 2)}{3} = x + 2$$

The factor 3 multiplies the entire numerator.

**2 Check with a number.**

At  $x = 2$ :  $\frac{x + 3}{x} = \frac{5}{2}$ , not 3.



## Real-world connection: Movie tickets

Suppose  $n$  movie tickets cost \$12 each, plus one \$6 booking fee per order.

$$\text{Average cost per ticket} = \frac{12n + 6}{n} = 12 + \frac{6}{n}$$

$$\text{For 3 tickets: } 12 + \frac{6}{3} = \$14 \text{ per ticket.}$$

Here,  $n$  is a positive whole number.



### MATH HINT

Ask: Does this factor multiply the ENTIRE numerator and denominator?

## Image 5 practice: Avoiding cancellation mistakes

**Q21.** A student claims  $\frac{3x + 12}{3x} = 4$ . Disprove the claim at  $x = 2$ , then give a correct expression and restriction.

**Q22.** A student simplifies  $\frac{6x^2 + 18x}{3x}$  to  $2x + 18$ . Correct the result and explain both cancellations.

**Q23.** Rewrite  $\frac{6x^2 - 9x + 12}{3x}$  as separate terms. State its restriction and evaluate it at  $x = -2$ .

**Q24.** A student says  $\frac{x^2 - 4x + 4}{x^2 - 4} = \frac{x - 2}{x + 2}$  for every  $x \neq -2$ . Correct the domain and compare the two formulas at  $x = 2$ .

**Q25.** A student cancels  $4 - x$  with  $x - 4$  in  $\frac{(4 - x)(x + 3)}{x^2 - 16}$  and gets  $\frac{x + 3}{x + 4}$ . Find and correct the sign error; state all exclusions.

# Multiplying Rational Expressions

Factor completely, cancel common factors, then multiply what remains.

$$\frac{x^2 - 4}{x + 1} \cdot \frac{x + 1}{x + 2}$$

**General rule:**

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

$$b \neq 0, d \neq 0$$

**1** Record the restrictions.

Original denominators:  $x + 1$  and  $x + 2$   
 $x \neq -1$  and  $x \neq -2$ .

**2** Factor and cancel.

$$\frac{(x - 2)(x + 2)}{x + 1} \cdot \frac{x + 1}{x + 2}$$

The common factors  $x + 1$  and  $x + 2$  cancel.

**3** Multiply the remaining factors.

$$x - 2$$

Restrictions remain:  $x \neq -1, -2$ .



## Real-world connection: Travel time

A train travels  $d$  kilometers at  $v$  kilometers per hour.

Time =  $\frac{d}{v}$  hours      Distance = speed  $\times$  time

$$v \cdot \frac{d}{v} = d$$

The factor  $v$  cancels. In this context,  $d > 0$  and  $v > 0$ .



### MATH HINT

Cancel common factors before multiplying to keep the algebra simpler.

**Image 6 practice: Multiplying rational expressions**

**Q26.** Simplify and state every excluded value:  $\frac{6x^2}{5} \cdot \frac{15}{9x}$ .

**Q27.** Simplify and state every excluded value:  $\frac{2x+8}{x-1} \cdot \frac{x^2-1}{4x+16}$ .

**Q28.** Simplify and state every excluded value:  $\frac{x^2-49}{2x+6} \cdot \frac{3x+9}{x^2+9x+14}$ .

**Q29.** Simplify and state every excluded value:  $\frac{2x^2+7x+3}{x^2-9} \cdot \frac{x-3}{4x+2}$ .

**Q30.** Simplify and state every excluded value:  $\frac{3x^2-12}{2x^2+8x+8} \cdot \frac{x^2+5x+6}{x^2-4x+4}$ .

# Dividing Rational Expressions

Multiply by the reciprocal of the divisor. The divisor must be defined and cannot equal zero.

$$\frac{x^2 - 9}{x + 2} \div \frac{x - 3}{x + 2}$$

**General rule:**

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c}$$

$$b \neq 0, c \neq 0, d \neq 0$$

## 1 Find all restrictions.

Denominators:  $x + 2 \neq 0 \rightarrow x \neq -2$ .

Divisor cannot be zero:  $x - 3 \neq 0 \rightarrow x \neq 3$ .

## 2 Multiply by the reciprocal.

$$\frac{x^2 - 9}{x + 2} \cdot \frac{x + 2}{x - 3}$$

Flip only the second fraction.

## 3 Factor, cancel, and simplify.

$$\frac{(x - 3)(x + 3)}{x + 2} \cdot \frac{x + 2}{x - 3} = x + 3$$

$x + 3$ , with  $x \neq -2$  and  $x \neq 3$ .



### Real-world connection: Filling bottles

You have 6 liters of water. Each bottle holds  $\frac{3}{4}$  liter.

$$\text{Number of bottles} = 6 \div \frac{3}{4} = 6 \times \frac{4}{3} = 8$$

Dividing by a fraction tells us how many equal portions fit.



#### MATH HINT

Check both original denominators and the numerator of the divisor before simplifying.

**Image 7 practice: Dividing rational expressions**

**Q31.** Simplify and state every excluded value:  $\frac{5x^2}{12} \div \frac{10x}{9}$ .

**Q32.** Simplify and state every excluded value:  $\frac{2x + 10}{x - 3} \div \frac{x + 5}{x^2 - 9}$ .

**Q33.** Simplify and state every excluded value:  $\frac{x^2 - 64}{2x + 6} \div \frac{x - 8}{x^2 + 6x + 9}$ .

**Q34.** Simplify and state every excluded value:  $\frac{2x^2 + 9x + 10}{x^2 - 4} \div \frac{2x + 5}{x^2 - 2x}$ .

**Q35.** Simplify and state every excluded value:  $\frac{3x^2 - 27}{x^2 + 4x + 3} \div \frac{2x - 6}{x^2 + 5x + 4}$ .

# Combining Multiplication and Division

Rewrite division as multiplication by a reciprocal, factor, and simplify.

$$\frac{x^2 - 1}{x} \cdot \frac{x}{x + 2} \div \frac{x - 1}{x + 2}$$

## 1 Record every restriction.

Original denominators:  $x \neq 0$  and  $x \neq -2$ .  
The divisor cannot be zero:  $x \neq 1$ .

## 2 Rewrite division and factor.

$$\frac{(x - 1)(x + 1)}{x} \cdot \frac{x}{x + 2} \cdot \frac{x + 2}{x - 1}$$

Take the reciprocal of the divisor only.

## 3 Cancel common factors.

Cancel  $x$ ,  $x + 2$ , and  $x - 1$ .

$$x + 1$$

Restrictions remain:  $x \neq 0, -2, 1$ .

Check at  $x = 3$ :  $\frac{8}{3} \cdot \frac{3}{5} \div \frac{2}{5} = 4$ .



### Real-world connection: Scaling a recipe

One batch uses  $\frac{3}{4}$  cup of flour and makes 6 servings. How much flour is needed for 10 servings?

$$\frac{3}{4} \times 10 \div 6 = \frac{3}{4} \times \frac{10}{6} = \frac{5}{4} \text{ cups}$$

That is  $1\frac{1}{4}$  cups of flour.



#### MATH HINT

Multiplication and division have equal priority. Work left to right, or rewrite division using reciprocals first.

**Image 8 practice: Combining multiplication and division**

**Q36.** Simplify and state every excluded value:  $\frac{4x^2}{9} \cdot \frac{15}{2x} \div \frac{5x}{6}$ .

**Q37.** Simplify and state every excluded value:  $\frac{2x+6}{x-4} \cdot \frac{x^2-16}{x+1} \div \frac{x+3}{2x+2}$ .

**Q38.** Simplify and state every excluded value:  $\frac{x^2-49}{2x} \cdot \frac{3x}{x+2} \div \frac{x-7}{2x+4}$ .

**Q39.** Simplify and state every excluded value:  $\frac{2x^2+11x+12}{x^2-16} \cdot \frac{x-4}{x+6} \div \frac{2x+3}{3x+18}$ .

**Q40.** Simplify and state every excluded value:  $\frac{x^2-36}{x^2-4} \div \frac{2x-12}{x+2} \cdot \frac{x^2-4x+4}{x+5}$ .

## Homework review: R1–R5

**R1.** For  $G(x) = \frac{2x^2 - 3x - 2}{2x + 1}$ , find the excluded value and evaluate  $G(-1)$  and  $G(2)$ .

**R2.** Find all exclusions of  $\frac{x - 3}{x(2x^2 - 5x - 3)}$ . Would canceling a common factor ever restore an excluded input?

**R3.** Simplify and state every excluded value:  $\frac{3x^2 + 18x + 24}{2x^2 + 4x}$ .

**R4.** Simplify and state every excluded value:  $\frac{4x^2 - 36}{2x^2 - 12x + 18}$ .

**R5.** A student claims  $\frac{5 - x}{x^2 - 25} = \frac{1}{x + 5}$ . Correct the sign and restrictions, then evaluate at  $x = 0$ .

## Homework review: R6–R8

**R6.** Simplify and state every excluded value:  $\frac{2x^2 - 50}{x^2 + 6x + 5} \cdot \frac{x + 1}{4x - 20}$ .

**R7.** Simplify and state every excluded value:  $\frac{x^2 + 9x + 20}{2x^2 - 32} \div \frac{x + 5}{x^2 - 3x - 4}$ .

**R8.** Simplify and state every excluded value:  $\frac{3x^2 - 48}{x^2 - 1} \cdot \frac{x - 1}{2x + 6} \div \frac{x - 4}{x^2 + 4x + 3}$ .

## Homework word problems: W1–W5

**W1.** A pottery club pays \$120 plus \$9 per participant, then receives a \$30 group discount. Write the average cost for  $n$  participants. How much less does each person pay with 18 participants than with 12?

**W2.** A cyclist rides 24 miles to a park in  $t$  hours and returns along the same route in  $t + \frac{1}{2}$  hours. Write the average speed for the entire trip and evaluate it when  $t = \frac{3}{2}$ . State a practical domain.

**W3.** A rectangular mat has area  $2x^2 + 13x + 15$  square meters and width  $2x + 3$  meters, with  $x > 0$ . Find its length in simplest form and its perimeter when  $x = 4$ .

**W4.** A display has remaining area  $4x^2 - 25$  square meters after a panel is removed. It is rearranged into a rectangle of width  $2x - 5$  meters, with  $x > \frac{5}{2}$ . Find its length and a simplified formula for its perimeter.

**W5.** A print shop charges  $12n + 60$  dollars for  $n$  posters and applies a \$24 order discount. A student says the average cost is always \$48. Correct the claim and compare the averages for 9 and 18 posters.

## Homework word problems: W6–W10

**W6.** A sign has length  $\frac{x^2 + 7x + 10}{x^2 - 1}$  meters and width  $\frac{3(x - 1)}{x + 2}$  meters, where  $x > 1$ .

Simplify its area, state all algebraic exclusions, and find the area at  $x = 3$ .

**W7.** A machine produces  $\frac{6x}{x + 4}$  parts per minute for  $\frac{x^2 + 7x + 12}{2x}$  minutes, where  $x$  is a positive whole number. Simplify the number produced. At  $x = 5$ , how many usable parts remain if three are rejected?

**W8.** A ribbon is  $\frac{2x^2 - 72}{x + 2}$  meters long. Each piece is  $\frac{x - 6}{x + 2}$  meter long, where  $x$  is a whole number greater than 6. Find the number of pieces, then the number available at  $x = 8$  after four pieces are reserved.

**W9.** A tank contains  $\frac{2x^2 + 11x + 12}{x + 1}$  liters. Each container holds  $\frac{2x + 3}{2x + 2}$  liters, with  $x$  a positive whole number. How many containers can be filled? At  $x = 2$ , how many are left for sale after one is kept as a sample?

**W10.** A panel has area  $\frac{x^2 - 16}{x + 5}$  square meters and width  $\frac{2x - 8}{x + 2}$  meters, with  $x > 4$ . Simplify its length and evaluate at  $x = 6$ . Explain why  $x = 4$  is forbidden even if it disappears during cancellation.

## Homework word problems: W11–W12

**W11.** A recipe uses  $\frac{x^2 + 8x + 15}{x + 1}$  cups of flour for  $2(x + 3)$  servings, with  $x$  a positive whole number. At the same rate, simplify the flour needed for  $3(x + 1)$  servings and evaluate at  $x = 3$ .

**W12.** A machine produces  $\frac{2x^2 - 50}{x + 1}$  meters of ribbon per minute for  $\frac{x^2 + 4x + 3}{3(x + 2)}$  minutes. Each spool holds  $\frac{x^2 - 25}{x + 2}$  meters, with  $x > 5$ . Simplify the number of spool-capacities produced. At  $x = 7$ , how many full spools and what fraction of another can be filled?